

Import Required Libraries

```
import pandas as pd
```

```
import numpy as np
```

Read Data From different formats

```
df = pd.read_csv("usa_housing_kaggle.csv")  
df.head()
```

	Price	Bedrooms	Bathrooms	SquareFeet	YearBuilt	GarageSpaces	LotSize
0	221958	1	1.9	4827	1979	2	1.45
1	771155	2	2.0	1035	1987	2	1.75
2	231932	1	3.0	2769	1982	1	1.46
3	465838	3	3.3	2708	1907	3	1.62
4	359178	4	3.4	1175	1994	2	0.74

Next steps:

[Generate code with df](#)[New interactive sheet](#)

Find Shape of Data

```
df.shape
```

```
(300, 10)
```

Find Missing Values

```
df.isna().sum()
```

	0
Price	0
Bedrooms	0
Bathrooms	0
SquareFeet	0
YearBuilt	0
GarageSpaces	0
LotSize	0
ZipCode	0
CrimeRate	0
SchoolRating	0

dtype: int64

Find data type of each column

df.dtypes

	0
Price	int64
Bedrooms	int64
Bathrooms	float64
SquareFeet	int64
YearBuilt	int64
GarageSpaces	int64
LotSize	float64
ZipCode	int64
CrimeRate	float64
SchoolRating	int64

dtype: object

Finding out Zero's

(df==0).sum()

	0
Price	0
Bedrooms	0
Bathrooms	0
SquareFeet	0
YearBuilt	0
GarageSpaces	72
LotSize	0
ZipCode	0
CrimeRate	0
SchoolRating	0

dtype: int64

Indexing and selecting data, sort data,

```
df['Price'].head()
```

	Price
0	221958
1	771155
2	231932
3	465838
4	359178

dtype: int64

```
df.columns
```

```
Index(['Price', 'Bedrooms', 'Bathrooms', 'SquareFeet', 'YearBuilt',  
      'GarageSpaces', 'LotSize', 'ZipCode', 'CrimeRate', 'SchoolRating'],  
      dtype='object')
```

Describe attributes of data

```
df.describe()
```

	Price	Bedrooms	Bathrooms	SquareFeet	YearBuilt	GarageSp
count	300.000000	300.000000	300.000000	300.000000	300.000000	300.00
mean	522761.916667	2.966667	2.464000	3052.750000	1962.616667	1.51
std	256922.348275	1.451175	0.886184	1194.97144	36.005431	1.11
min	112666.000000	1.000000	1.000000	840.000000	1902.000000	0.00
25%	299936.500000	2.000000	1.700000	2053.500000	1931.750000	1.00
50%	505806.500000	3.000000	2.400000	3127.000000	1961.000000	1.00
75%	744603.750000	4.000000	3.200000	4081.000000	1994.000000	3.00
max	999684.000000	5.000000	4.000000	4997.000000	2022.000000	3.00

Counting unique values of data

```
df.nunique()
```

	0
Price	300
Bedrooms	5
Bathrooms	31
SquareFeet	285
YearBuilt	111
GarageSpaces	4
LotSize	154
ZipCode	300
CrimeRate	295
SchoolRating	9

dtype: int64

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Read Dataframe

Read Dataframe

```
d=pd.DataFrame({
  'ID':[101, 102, 103 ,104, 105],
  'NAME': ['Prajakta', 'Rutika', 'Srushti' , 'Pooja', 'Bhoomi'],
  'AGE': [19, 19, 19, 19, 19],
  'GENDER':['Female', 'Female', 'Female', 'Female', 'Female'],
```

```
'DEPT': ['CSE-AI', 'IT' , 'IT' , 'IT' , 'CSE-AI'],
'MARKS': [93 ,92 ,90, 91, 92],
'ATTENDANCE': [95 ,92, 91, 95 ,95],
'INTERSHIP': ['Yes' , 'Yes' , 'Yes' , 'NO' , 'Yes']
})
```

```
print(d)
```

	ID	NAME	AGE	GENDER	DEPT	MARKS	ATTENDANCE	INTERSHIP
0	101	Prajakta	19	Female	CSE-AI	93	95	Yes
1	102	Rutika	19	Female	IT	92	92	Yes
2	103	Srushti	19	Female	IT	90	91	Yes
3	104	Pooja	19	Female	IT	91	95	NO
4	105	Bhoomi	19	Female	CSE-AI	92	95	Yes

Viewing Data

```
d.head()
```

	ID	NAME	AGE	GENDER	DEPT	MARKS	ATTENDANCE	INTERSHIP	
0	101	Prajakta	19	Female	CSE-AI	93	95	Yes	
1	102	Rutika	19	Female	IT	92	92	Yes	
2	103	Srushti	19	Female	IT	90	91	Yes	
3	104	Pooja	19	Female	IT	91	95	NO	
4	105	Bhoomi	19	Female	CSE-AI	92	95	Yes	

Next steps:

[Generate code with d](#)[New interactive sheet](#)

```
d.tail(3)
```

	ID	NAME	AGE	GENDER	DEPT	MARKS	ATTENDANCE	INTERSHIP	
2	103	Srushti	19	Female	IT	90	91	Yes	
3	104	Pooja	19	Female	IT	91	95	NO	
4	105	Bhoomi	19	Female	CSE-AI	92	95	Yes	

```
d.index
```

```
RangeIndex(start=0, stop=5, step=1)
```

```
d.columns
```

```
Index(['ID', 'NAME', 'AGE', 'GENDER', 'DEPT', 'MARKS', 'ATTENDANCE',  
      'INTERSHIP'],  
      dtype='object')
```

```
d.to_numpy()
```

```
array([[101, 'Prajakta', 19, 'Female', 'CSE-AI', 93, 95, 'Yes'],  
       [102, 'Rutika', 19, 'Female', 'IT', 92, 92, 'Yes'],  
       [103, 'Srushti', 19, 'Female', 'IT', 90, 91, 'Yes'],  
       [104, 'Pooja', 19, 'Female', 'IT', 91, 95, 'NO'],  
       [105, 'Bhoomi', 19, 'Female', 'CSE-AI', 92, 95, 'Yes']],  
      dtype=object)
```

```
d.dtypes
```

	0
ID	int64
NAME	object
AGE	int64
GENDER	object
DEPT	object
MARKS	int64
ATTENDANCE	int64
INTERSHIP	object

```
dtype: object
```

```
d.describe()
```

	ID	AGE	MARKS	ATTENDANCE	
count	5.000000	5.0	5.000000	5.000000	
mean	103.000000	19.0	91.600000	93.600000	
std	1.581139	0.0	1.140175	1.949359	
min	101.000000	19.0	90.000000	91.000000	
25%	102.000000	19.0	91.000000	92.000000	
50%	103.000000	19.0	92.000000	95.000000	
75%	104.000000	19.0	92.000000	95.000000	
max	105.000000	19.0	93.000000	95.000000	

```
d.sort_index(axis=1, ascending=False)
```

	NAME	MARKS	INTERSHIP	ID	GENDER	DEPT	ATTENDANCE	AGE	
0	Prajakta	93	Yes	101	Female	CSE-AI	95	19	
1	Rutika	92	Yes	102	Female	IT	92	19	
2	Srushti	90	Yes	103	Female	IT	91	19	
3	Pooja	91	NO	104	Female	IT	95	19	
4	Bhoomi	92	Yes	105	Female	CSE-AI	95	19	

```
d.sort_values(by="MARKS", ascending=False)
```

	ID	NAME	AGE	GENDER	DEPT	MARKS	ATTENDANCE	INTERSHIP	
0	101	Prajakta	19	Female	CSE-AI	93	95	Yes	
1	102	Rutika	19	Female	IT	92	92	Yes	
4	105	Bhoomi	19	Female	CSE-AI	92	95	Yes	
3	104	Pooja	19	Female	IT	91	95	NO	
2	103	Srushti	19	Female	IT	90	91	Yes	

Selection

Getitem[]

```
d["MARKS"]
```

	MARKS
0	93
1	92
2	90
3	91
4	92

dtype: int64

```
d[["NAME", "MARKS"]]
```

	NAME	MARKS
0	Prajakta	93
1	Rutika	92
2	Srushti	90
3	Pooja	91
4	Bhoomi	92

```
d[0:3]
```

	ID	NAME	AGE	GENDER	DEPT	MARKS	ATTENDANCE	INTERSHIP
0	101	Prajakta	19	Female	CSE-AI	93	95	Yes
1	102	Rutika	19	Female	IT	92	92	Yes
2	103	Srushti	19	Female	IT	90	91	Yes

Selection by Lable

```
d.loc[:, ["NAME", "MARKS"]]
```


	NAME	MARKS
0	Prajakta	93
1	Rutika	92
2	Srushti	90
3	Pooja	91
4	Bhoomi	92

```
d.loc[0, "NAME"]
```

```
'Prajakta'
```

```
d.at[0, "MARKS"]
```

```
np.int64(93)
```

Selection by position -- `iloc[]`

```
d.iloc[3]
```

	3
ID	104
NAME	Pooja
AGE	19
GENDER	Female
DEPT	IT
MARKS	91
ATTENDANCE	95
INTERSHIP	NO

```
dtype: object
```

```
d.iloc[1:4, 0:3]
```

	ID	NAME	AGE	
1	102	Rutika	19	
2	103	Srushti	19	
3	104	Pooja	19	

```
d.iloc[[0,2,4],[1,5]]
```

	NAME	MARKS	
0	Prajakta	93	
2	Srushti	90	
4	Bhoomi	92	

Boolean Indexing

```
d[d["MARKS"]>91]
```

	ID	NAME	AGE	GENDER	DEPT	MARKS	ATTENDANCE	INTERSHIP	
0	101	Prajakta	19	Female	CSE-AI	93	95	Yes	
1	102	Rutika	19	Female	IT	92	92	Yes	
4	105	Bhoomi	19	Female	CSE-AI	92	95	Yes	

```
d[d["DEPT"].isin(["CSE-AI"])]
```

	ID	NAME	AGE	GENDER	DEPT	MARKS	ATTENDANCE	INTERSHIP	
0	101	Prajakta	19	Female	CSE-AI	93	95	Yes	
4	105	Bhoomi	19	Female	CSE-AI	92	95	Yes	

Setting Values

```
d.at[0,"MARKS"]=95
```

```
d
```

	ID	NAME	AGE	GENDER	DEPT	MARKS	ATTENDANCE	INTERSHIP	
0	101	Prajakta	19	Female	CSE-AI	95	95	Yes	
1	102	Rutika	19	Female	IT	92	92	Yes	
2	103	Srushti	19	Female	IT	90	91	Yes	
3	104	Pooja	19	Female	IT	91	95	NO	
4	105	Bhoomi	19	Female	CSE-AI	92	95	Yes	

Next steps:

[Generate code with d](#)[New interactive sheet](#)

```
d.iat[1, 6] = 96
```

d

	ID	NAME	AGE	GENDER	DEPT	MARKS	ATTENDANCE	INTERSHIP	
0	101	Prajakta	19	Female	CSE-AI	95	95	Yes	
1	102	Rutika	19	Female	IT	92	96	Yes	
2	103	Srushti	19	Female	IT	90	91	Yes	
3	104	Pooja	19	Female	IT	91	95	NO	
4	105	Bhoomi	19	Female	CSE-AI	92	95	Yes	

Next steps:

[Generate code with d](#)[New interactive sheet](#)

```
s1=pd.Series([1,2,3,4,5])
d["GRADE"]=s1
```

d

	ID	NAME	AGE	GENDER	DEPT	MARKS	ATTENDANCE	INTERSHIP	GRADE
0	101	Prajakta	19	Female	CSE-AI	95	95	Yes	1
1	102	Rutika	19	Female	IT	92	96	Yes	2
2	103	Srushti	19	Female	IT	90	91	Yes	3
3	104	Pooja	19	Female	IT	91	95	NO	4
4	105	Bhoomi	19	Female	CSE-AI	92	95	Yes	5



Next steps:

[Generate code with d](#)[New interactive sheet](#)

Operations Stats(Mean)

```
d.mean(numeric_only=True)
```

	0
ID	103.0
AGE	19.0
MARKS	91.6
ATTENDANCE	93.6

dtype: float64

```
d["MARKS"].mean()
```

```
np.float64(91.6)
```

User defined functions

```
d[["MARKS", "ATTENDANCE"]].agg(lambda x: x.mean() * 2)
```

	0
MARKS	183.2
ATTENDANCE	187.2

dtype: float64

```
d["MARKS"].transform(lambda x: x + 5)
```

	MARKS
0	98
1	97
2	95
3	96
4	97

dtype: int64

Value Counts

```
d["DEPT"].value_counts()
```

	count
DEPT	
IT	3
CSE-AI	2

dtype: int64

String Methods

```
d["NAME"].str.lower()
```

	NAME
0	prajakta
1	rutika
2	srushti
3	pooja
4	bhoomi

dtype: object

Merge

Concat

```
result = pd.DataFrame({'RESULT':['Pass','Pass','Pass','Pass','Pass']})
```

```
pd.concat([d, result], axis=1)
```

	ID	NAME	AGE	GENDER	DEPT	MARKS	ATTENDANCE	INTERSHIP	RESULT
0	101	Prajakta	19	Female	CSE-AI	93	95	Yes	Pass
1	102	Rutika	19	Female	IT	92	92	Yes	Pass
2	103	Srushti	19	Female	IT	90	91	Yes	Pass
3	104	Pooja	19	Female	IT	91	95	NO	Pass
4	105	Bhoomi	19	Female	CSE-AI	92	95	Yes	Pass

Join

```
dept = pd.DataFrame({
    'DEPT':['CSE-AI','IT'],
    'HOD':['Dr. Sharma','Dr. Patil']
})
```

```
pd.merge(d, dept, on='DEPT')
```

	ID	NAME	AGE	GENDER	DEPT	MARKS	ATTENDANCE	INTERSHIP	HOD
0	101	Prajakta	19	Female	CSE-AI	93	95	Yes	Dr. Sharma
1	102	Rutika	19	Female	IT	92	92	Yes	Dr. Patil
2	103	Srushti	19	Female	IT	90	91	Yes	Dr. Patil
3	104	Pooja	19	Female	IT	91	95	NO	Dr. Patil
4	105	Bhoomi	19	Female	CSE-AI	92	95	Yes	Dr. Sharma

```
d.sort_values(by='ATTENDANCE', ascending=True)
```

	ID	NAME	AGE	GENDER	DEPT	MARKS	ATTENDANCE	INTERSHIP
2	103	Srushti	19	Female	IT	90	91	Yes
1	102	Rutika	19	Female	IT	92	92	Yes
0	101	Prajakta	19	Female	CSE-AI	93	95	Yes
3	104	Pooja	19	Female	IT	91	95	NO
4	105	Bhoomi	19	Female	CSE-AI	92	95	Yes

```
d.isnull()
```

	ID	NAME	AGE	GENDER	DEPT	MARKS	ATTENDANCE	INTERSHIP
0	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False

```
d.isna()
```

	ID	NAME	AGE	GENDER	DEPT	MARKS	ATTENDANCE	INTERSHIP
0	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False

```
d.notna()
```

	ID	NAME	AGE	GENDER	DEPT	MARKS	ATTENDANCE	INTERSHIP
0	True	True	True	True	True	True	True	True
1	True	True	True	True	True	True	True	True
2	True	True	True	True	True	True	True	True
3	True	True	True	True	True	True	True	True
4	True	True	True	True	True	True	True	True

```
d.dropna()
```

	ID	NAME	AGE	GENDER	DEPT	MARKS	ATTENDANCE	INTERSHIP
0	101	Prajakta	19	Female	CSE-AI	93	95	Yes
1	102	Rutika	19	Female	IT	92	92	Yes
2	103	Srushti	19	Female	IT	90	91	Yes
3	104	Pooja	19	Female	IT	91	95	NO
4	105	Bhoomi	19	Female	CSE-AI	92	95	Yes

```
d.fillna(value=5)
```

	ID	NAME	AGE	GENDER	DEPT	MARKS	ATTENDANCE	INTERSHIP
0	101	Prajakta	19	Female	CSE-AI	93	95	Yes
1	102	Rutika	19	Female	IT	92	92	Yes
2	103	Srushti	19	Female	IT	90	91	Yes
3	104	Pooja	19	Female	IT	91	95	NO
4	105	Bhoomi	19	Female	CSE-AI	92	95	Yes

```
d.interpolate()
```

```
/tmp/ipython-input-3264332000.py:1: FutureWarning: DataFrame.interpolate with d.interpolate()
```

	ID	NAME	AGE	GENDER	DEPT	MARKS	ATTENDANCE	INTERSHIP
0	101	Prajakta	19	Female	CSE-AI	93	95	Yes
1	102	Rutika	19	Female	IT	92	92	Yes
2	103	Srushti	19	Female	IT	90	91	Yes
3	104	Pooja	19	Female	IT	91	95	NO
4	105	Bhoomi	19	Female	CSE-AI	92	95	Yes

```
d.loc[len(d)] = [102, 'Rutika', 19, 'Female', 'IT', 92, 92, 'Yes']
```

```
d
```


	ID	NAME	AGE	GENDER	DEPT	MARKS	ATTENDANCE	INTERSHIP
0	101	Prajakta	19	Female	CSE-AI	93	95	Yes
1	102	Rutika	19	Female	IT	92	92	Yes

Double-click (or Enter) to edit

2	103	Srushti	19	Female	IT	90	91	Yes
3	104	Pooja	19	Female	IT	91	95	NO

d.duplicated()

5	102	Rutika	19	Female	IT	92	92	Yes
---	-----	--------	----	--------	----	----	----	-----

0 False

1 False

2 False

3 False

4 False

5 True

dtype: bool

d.drop_duplicates()

	ID	NAME	AGE	GENDER	DEPT	MARKS	ATTENDANCE	INTERSHIP
0	101	Prajakta	19	Female	CSE-AI	93	95	Yes
1	102	Rutika	19	Female	IT	92	92	Yes
2	103	Srushti	19	Female	IT	90	91	Yes
3	104	Pooja	19	Female	IT	91	95	NO
4	105	Bhoomi	19	Female	CSE-AI	92	95	Yes

d['AGE']=d['AGE'].astype(float)

print(d.dtypes)

```
ID          int64
NAME        object
AGE         float64
GENDER      object
DEPT        object
MARKS       int64
ATTENDANCE  int64
INTERSHIP   object
dtype: object
```

```
d.convert_dtypes
```

```
pandas.core.generic.NDFrame.convert_dtypes
```

```
def convert_dtypes(infer_objects: bool_t=True, convert_string:
bool_t=True, convert_integer: bool_t=True, convert_boolean: bool_t=True,
convert_floating: bool_t=True, dtype_backend:
DtypeBackend='numpy_nullable') -> Self
```

```
e          Int64
```

```
f          Float64
```

```
dtype: object
```

Start with a Series of strings and missing data represented by ``np.nan``.

```
d
```

	ID	NAME	AGE	GENDER	DEPT	MARKS	ATTENDANCE	INTERSHIP
0	101	Prajakta	19.0	Female	CSE-AI	93	95	Yes
1	102	Rutika	19.0	Female	IT	92	92	Yes
2	103	Srushti	19.0	Female	IT	90	91	Yes
3	104	Pooja	19.0	Female	IT	91	95	NO
4	105	Bhoomi	19.0	Female	CSE-AI	92	95	Yes
5	102	Rutika	19.0	Female	IT	92	92	Yes

```
d.shape
```

```
(5, 8)
```